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22MA201
ENGINEERING MATHEMATICS II

UNIT IV
VECTOR INTEGRAL CALCULUS

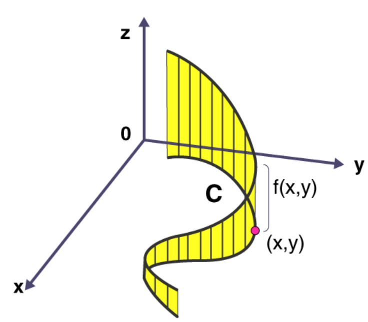
1. LINE INTEGRALS OF VECTOR POINT FUNCTIONS

1.1 LINE INTEGRALS OF VECTOR POINT FUNCTIONS

A line integral (sometimes called a path integral) is the integral of some function along a curve. One can integrate a scalar-valued function along a curve, obtaining for example, the mass of a wire from its density.

1.1.1 INTRODUCTION

A line integral is integral in which the function to be integrated is determined along a curve in the coordinate system. The function which is to be integrated may be either a scalar field or a vector field. We can integrate a scalar-valued function or vector-valued function along a curve. The value of the line integral can be evaluated by adding all the values of points on the vector field.



1.1.2 DEFINITION

If $\vec{F}$ represents the force acting on a particle moving along an arc AB then the work done during the small displacement $\delta \vec{R} = \vec{F} . \delta \vec{R}$. The total work done by $\vec{F}$ during the displacement from A to B is given by the line integral $\int_A^B \vec{F} . d\vec{R}$.

1.1.3 PROBLEMS

Example 1:

If $\vec{F} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$ evaluate $\int \vec{F} \cdot d\vec{R}$ from (0, 0, 0) to (1, 1, 1) along the path $x = t, y = t^2, z = t^3$

Solution:

Given, $\vec{F} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$

$$x = t, y = t^2, z = t^3$$

$$dx = dt \; dy = 2tdt \; dz = 3t^2dt$$

$$d\vec{R} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\begin{aligned} \vec{F} \cdot d\vec{R} &= (3x^2 + 6y)dx - 14yzdy + 20xz^2dz \\ &= (3t^2 + 6t^2)dt - 14(t^2)t^3 (2tdt) + 20(t)(t^3)^2(3t^2dt) \\ &= [9t^2 - 28t^6 + 60\,t^8]dt \end{aligned}$$

t varies from 0 to 1

$$\begin{aligned} \int \vec{F} \cdot d\vec{R} &= \int_0^1 [9t^2 - 28t^6 + 60t^9]dt \\ &= \left[\frac{9t^3}{3} - \frac{28t^7}{7} + \frac{60t^{10}}{10}\right]_0^1 \\ &= [3t^3 - 4t^7 + 6t^{10}]_0^1 \\ &= 3 - 4 + 6 - 0 \\ \int \vec{F} \cdot d\vec{R} &= 5 \end{aligned}$$

Example 2:

Find the work done by a force $\vec{F} = (x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}$ which moves a particle in xy plane from (0, 0) to (1, 1) along the parabola $y^2 = x$. Is the work done different when the path is the straight line $y = x$.

Solution:

Given, $\vec{F} = (x^2 - y^2 + x)\,\vec{i} - (2xy + y)\,\vec{j}$

$d\vec{R} = dx\vec{i} + dy\vec{j} + dz\vec{k}$

Given, $y^2 = x \Rightarrow 2y\,dy = dx$

Work done is

$$\int_c \vec{F}\cdot d\vec{R} = \int_c (x^2 - y^2 + x)dx - (2xy + y)dy$$

$$= \int_c ((y^2)^2 - y^2 + y^2)\,2ydy - (2y^2y + y)dy$$

$$= \int_c (y^4)2ydy - (2y^3 + y)dy$$

$$= \int_c [2y^5 - 2y^3 - y]dy$$

y varies from 0 to 1

$$\int_c \vec{F}\cdot d\vec{R} = \int_0^1 (2y^5 - 2y^3 - y)dy$$

$$= \left[\frac{2y^6}{6} - \frac{2y^4}{4} - \frac{y^2}{2}\right]_0^1$$

$$= \frac{1}{3} - \frac{1}{2} - \frac{1}{2} - 0$$

$$= \frac{1}{3} - 1$$

$$\int_c \vec{F}\cdot d\vec{R} = -\frac{2}{3}$$

If $x = y$

$dy = dx$

Then

$$\int_c \vec{F}\cdot d\vec{R} = \int_c (x^2 - x^2 + x)dx - (2x^2 + x)dx$$

$$= \int_c (x - 2x^2 - x)dx = \int_c -2x^2dx$$

x varies from 0 to 1

$$= \int_0^1 -2x^2dx$$

$$= \left[-\frac{2x^3}{3}\right]_0^1$$

Work done $\int_c \vec{F} \cdot d\vec{R} = -\frac{2}{3}$

Hence the work done is same when the path is a straight line $y = x$ and $y = x^2$.

Example 3:

Find the work done in moving a particle in the force field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line from (0, 0, 1) to (2, 1, 3).

Solution:

$$\text{Work done} = \int_c \vec{F} \cdot d\vec{R}$$

Given, $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$

Equation of line is

$$\frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-1}{3-1}$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z-1}{2} = t(\text{say})$$

$$\begin{aligned} x &= 2t \qquad & dx &= 2dt \\ y &= t \qquad & dy &= dt \\ z &= 2t - 1 \qquad & dz &= 2dt \end{aligned}$$

$$\int_c \vec{F} \cdot d\vec{R} = \int_c 3x^3 dx + (2xz - y)dy + zdz$$

$$= \int_c 3(2t)^2(2dt) + (2(2t)(2t-1) - t)dt + (2t-1)2dt$$

$$= \int_c [24t^2 + 8t^2 - 4t - t + 4t - 2]dt$$

$$= \int_c [32t^2 - t - 2]dt$$

t varies from 0 to 1

$$= \int_0^1 (32t^2 - t - 2)dt$$

$$= \left[\frac{32t^3}{3} - \frac{t^2}{2} - 2t\right]_0^1$$

$$= \frac{32}{3} - \frac{1}{2} - 2$$

$$= \frac{64-3-12}{6} = \frac{49}{6}$$

$$\text{Work done} = \frac{49}{6}$$

Example 4:

If $\vec{F} = 3xy\vec{i} - y^2\vec{j}$ Evaluate $\int \vec{F}\cdot d\vec{R}$ where c is the curve in the plane $y = 2x^2$ from $(0, 0)$ to $(1, 2)$.

Solution:

Given, $\vec{F} = 3xy\vec{i} - y^2\vec{j}$, $y = 2x^2$

$$d\vec{R} = dx\vec{i} + dy\vec{j}$$

$$\vec{F}\cdot d\vec{R} = (3xy\vec{i} - y^2\vec{j})\cdot(dx\vec{i} + dy\,\vec{j})$$

$$= 3xy\,dx - y^2dy$$

Put $y = 2x^2$

$dy = 4xdx$

$$\vec{F}\cdot d\vec{R} = 3x(2x^2)dx - (2x^2)^2(4xdx)$$

$$= 6x^3dx - 16x^5dx$$

$$\int_c \vec{F}\cdot d\vec{R} = \int_c (6x^3 - 16x^5)dx$$

x varies from 0 to 1

$$= \int_0^1 (6x^3 - 16x^5)dx = \left[\frac{6x^4}{4} - \frac{16x^6}{6}\right]_0^1$$

$$= \left[\frac{3}{2}x^4 - \frac{8x^6}{3}\right]_0^1 = \left[\frac{3}{2} - \frac{8}{3} - 0\right]$$

$$\int_c \vec{F}\cdot d\vec{R} = \left[\frac{9-16}{6}\right]$$

$$\int_c \vec{F}\cdot d\vec{R} = -\frac{7}{6}$$

Example 5:

Compute the line integral $\int_c (y^2dx - x^2dy)$ about the triangle whose vertices are (1, 0) (0, 1) and (−1, 0).

Solution:

Given, $\int (y^2dx - x^2dy)$

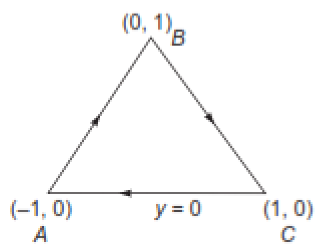
Vertices are (1, 0) (0, 1) and (−1, 0)

Equation of AB is

$$\frac{x+1}{0+1} = \frac{y-0}{1-0}$$

$$x + 1 = y - 0$$

$$\boxed{x = y-1} \qquad dx = dy$$



Equation of CA is $\boxed{y=0}$ $\qquad dy = 0$

Equation of BC is $\frac{x-0}{1-0} = \frac{y-1}{0-1} \Rightarrow x = -y + 1, y = 1 - x, dy = -dx$

In AB, $\int_{AB} y^2dx - x^2dy = \int_{AB} y^2\, dy - (y-1)^2dy = \int_{AB} (y^2 - y^2 - 1 + 2y)dy$

y varies from 0 to 1

$$= \int_0^1 (-1 + 2y)dy = \left[-y + \frac{\cancel{2}y^2}{\cancel{2}}\right]_0^1 = -1 + 1 = 0$$

In BC, $\int_{BC} y^2\,dx - x^2dy = \int_{BC} (1-x)^2dx = x^2(-dx) = \int_{BC} (1 + x^2 - 2x + x^2)dx$

x varies from 0 to 1

$$= \int_0^1 (2x^2 - 2x + 1)dx = \left[\frac{2x^3}{3} - \frac{2x^2}{2} + x\right]_0^1 = \frac{2}{3} - 1 + 1$$

$$= \frac{2}{3}$$

In CA, $\int_{CA} y^2dx - x^2dy = \int 0 = 0$ $\qquad [\,y = 0, dy = 0]$

$$\Rightarrow \qquad \int_c y^2dx - x^2dy = \int_{AB} + \int_{BC} + \int_{CA} = \frac{2}{3}$$

<u>Discussion Questions:</u>

1. Why is it called line integral?
2. What does a line integral represent in a vector field?
3. Is line integral a vector or scalar?
4. Is line integral always positive?
5. What are the application of line integral in vector field?